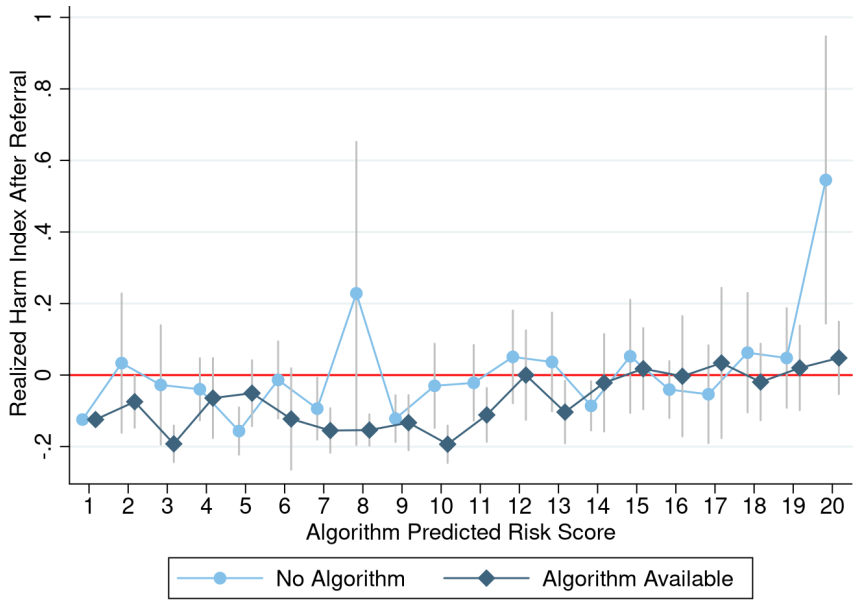
the reduction in harm for screened-out children on referrals scoring a 20 is statistically significant at the 90 percent confidence level ($\beta = -0.46, p = 0.070$). Hence, much of the capacity for harm reduction is already realized by social workers using the algorithm freely, such that further gains from a screen-in mandate would be comparatively modest, especially relative to the costs of additional investigations that would result from such a policy.

Figure 6 confirms that although workers did not change their screen-in rates for children with the highest algorithm risk scores when using the tool (Figure A.12), workers did screen in a set of ex-post more vulnerable children. Furthermore, in situations where workers "disagreed" with the algorithm (high algorithm score, worker screen-out), harm for high-score, screened out children was comparable with the control group harm mean (horizontal red line). Therefore, ex-post partial automation of investigation decisions using algorithm predictions alone is likely to be of limited benefit in our setting. Future work with larger study samples could consider partial automation with optimal triage, as proposed by Raghu et al. (2019).

Figure 6: Realized Harm for Screened-Out Children, by Algorithm Risk Score

Would a Default Screen-In Requirement Catch Vulnerable Children that Workers Screened Out?

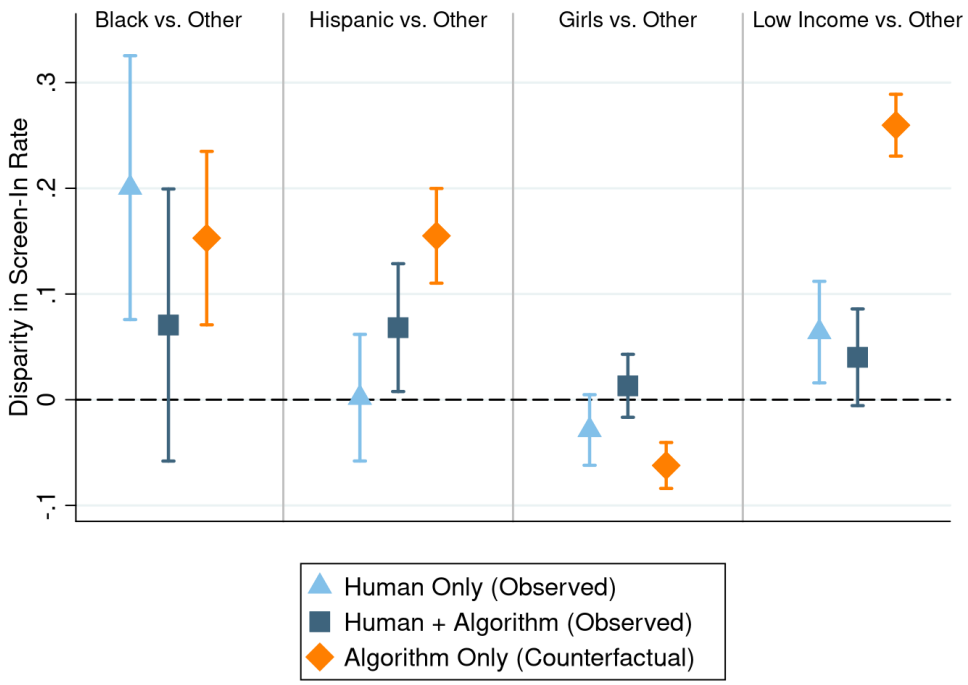


Notes: This figure shows the average realized harm index of children who were screened out, by treatment status and algorithm risk score. The dark blue plot shows the means when the tool is available. The light blue plot shows the means for the control group, where the tool is not available. 90% confidence intervals reported throughout. The horizontal red line is the average harm in the control group (for all children, both screened in and screened out).

6.3 Counterfactual 3: Algorithms and Disparities without Human Oversight

Finally, we estimate with minimal assumptions how an algorithm-only decision rule would have impacted screen-in equity. This is possible through the counterfactual algorithm screen-in decision rule described in Section 6.1, which only assumes no change in the overall screen-in rate and that the algorithm would have prioritized screen-ins for children with the highest predicted risk. Figure 7 shows CPS surveillance (screen-in) disparities for relevant demographic subgroups under three decision regimes: human-only, human-plus-algorithm, and algorithm-only.

Figure 7: Counterfactual: Deferring to Algorithm would Increase CPS Surveillance Disparities



Notes: This figure shows point estimates and 90 percent confidence intervals for disparities in screen-in rates by demographic category. Human-only and human-plus-algorithm disparities are directly observed in the data. Algorithm-only disparities are inferred with minimal assumptions: control group observations are sorted by algorithm risk score, and mandatory screen-ins are imposed above a score threshold that keeps overall screen-in rates constant. The vertical axis shows the disparity in screen-in rates. A zero point estimate represents zero disparity in screen-ins and is illustrated using a horizontal dashed black line. For example, in the first panel ("Black vs. Other"), screen-in rates were 20 percentage points greater for Black children compared to other children for decisions where humans did not have access to the algorithm ("Human Only"; light blue triangle on the left). Low income refers to children whose families are receiving SNAP benefits.

We find that human-plus-algorithm decisions improved equity not only compared to the status quo of human experts alone, but also compared to counterfactual screen-ins from the algorithm without human oversight. Relative to an algorithm-only decision rule, human oversight reduced disparities in CPS investigations by race, ethnicity, gender, and socioeconomic status: the dark blue squares in Figure 7 (human with

algorithm) are always closer to zero than the orange diamonds (algorithm only), though point estimates are in some cases imprecise. Second, the algorithm on its own would have investigated significantly more Hispanic and low-income children relative to the status quo of human experts on their own (light blue triangles). Increased surveillance under the algorithm-only counterfactual is not associated with a greater reduction in harm for these subgroups relative to observed human-plus-algorithm decisions (Appendix Figure A.17), suggesting that the algorithm-only disparities may not be warranted. In short, the improvements in equity observed during the trial would not have been fully realized – and for some groups may have worsened – without human oversight of the algorithm. Providing algorithm support to humans leads to more equitable levels of surveillance by CPS, relative to the algorithm on its own.

Estimating child health disparities under a counterfactual algorithm-only scenario requires additional assumptions about similar effects of screen-ins across subgroups. We present estimates in Appendix Figure A.18. Human-plus-algorithm decisions appear to be more effective at reducing health disparities than the algorithm on its own, though the differences are not statistically significant.

We conclude that human oversight of the algorithm for investigation decisions not only both improved efficiency and equity relative to a human-only scenario, but also improved equity relative to deferring all decisions to algorithms. Combining algorithm support with "in-the-loop" human expert oversight could be an effective strategy to overcome public skepticism about algorithms and the perpetuation of disparities.

7 Generalizability and External Validity

This section identifies decision environments where our findings are most likely to be informative. We develop a theoretical model that helps rationalize core results, and highlight two contextual features that may have been especially important given prior research. We conclude with a cost-benefit analysis.

7.1 Group Decision-Making

Group-based decision-making is common in high-stakes settings with large amounts of information and professional discretion: for example juries and parole boards, criminal justice and health care expert boards, and hiring committees. Similarly, as required by state legislation, child protection investigation decisions in the partner agency were made in small teams. Team decision-making may have contributed to the success of the algorithm tool in our setting. A longstanding literature on the "wisdom of the crowds" highlights how groups can often aggregate information more effectively than individuals (Galton, 1907). The literature on